

# NAPCO Nucleus

## Developer Setup Guide

*Nine steps. Run every command in PowerShell on your Windows dev PC.*

### START HERE

## Before you begin

Run every command below in **PowerShell** on your Windows dev PC. Total time: ~25 min, mostly waiting for `pip install`.

### Three things Titu DMs you first

Never share these on group chat — they contain secrets.

1. `.env` — save inside your repo folder (`$NN\.env`) after Step 1.
2. `google-credentials.json` — save inside your repo folder (`$NN\google-credentials.json`).
3. **This PDF.**

*The Samba password is inline in Step 5.*

### STEP 1

## Clone the repo

Pick install location. Set `$NN` to that path — every later step uses it.

Run in **PowerShell**:

```
$NN = "E:\Projects\NAPCO-Nucleus" # change if you want it elsewhere
mkdir (Split-Path $NN -Parent) -Force | Out-Null
git clone https://github.com/napco-labs/napco-nucleus.git $NN
Set-Location $NN
```

*`$NN` lives only in this PowerShell session. If you open a new shell later, set it again at the top. To persist across sessions:*

```
[Environment]::SetEnvironmentVariable("NN", $NN, "User")
```

*Then use `$env:NN` from any future shell.*

### STEP 2

## Install Python packages

Run in **PowerShell**:

```
Set-Location $NN
python -m pip install -r requirements.txt
```

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### STEP 3

## Install Tesseract OCR

Run anywhere in **PowerShell**:

```
winget install UB-Mannheim.TesseractOCR
```

Close and reopen PowerShell (re-set \$NN after reopen).

### STEP 4

## Place files + set your dev name

Save the two files Titu sent you into your repo folder:

```
$NN\.env  
$NN\google-credentials.json
```

Open .env in Notepad. Find this line:

```
NUCLEUS_DEV_NAME=Titu
```

Replace Titu with **your** name. Use one exactly:

```
Assad Rocky Ferdows Titu Atik Isruk Amin
```

Save. Close.

### STEP 5

## Cache Samba password (critical)

Use **REGULAR** PowerShell, **NOT** admin. The credential is stored per-user and must be in your normal session.

Run in **PowerShell**:

```
cmdkey /add:172.16.205.123 /user:nucleus /pass:E7CqJ0d1oHox7HTjxNp_osD_fSyUe59I  
Test-Path \\172.16.205.123\nucleus-central
```

**Expected:** Test-Path **must print** True. If it prints False, your calls won't upload to central — stop and ping Titu.

### STEP 6

## Install voice daemon

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Run in **PowerShell**:

```
Set-Location $NN
.\scripts\register-voice-daemon-task.ps1
Start-ScheduledTask -TaskName 'NAPCO Nucleus - Voice Daemon'
```

### STEP 7

## Install chat-push tasks

Run in **PowerShell**:

```
Set-Location $NN
.\scripts\register-chat-push-task.ps1
```

### STEP 8

## Enable remote ops (admin one-time)

So Titu can manage your PC from his desk without you needing to be at the keyboard. Open **admin PowerShell** (right-click PowerShell → Run as administrator):

```
Enable-PSRemoting -Force
Add-LocalGroupMember -Group "Remote Management Users" -Member "AEL\khasan"
```

If Enable-PSRemoting errors about network profile being "Public", use this instead:

```
Enable-PSRemoting -Force -SkipNetworkProfileCheck
Add-LocalGroupMember -Group "Remote Management Users" -Member "AEL\khasan"
```

If Add-LocalGroupMember says "User already member", that's fine — already granted.

### STEP 9

## Test

Make any Teams call (at least 20 seconds). Wait 2 minutes. Then:

```
$you = ((Select-String -Path "$NN\.env" -Pattern '^NUCLEUS_DEV_NAME=')).Line -replace
'NUCLEUS_DEV_NAME=', '').Trim()
$today = Get-Date -Format "yyyy-MM-dd"
Get-ChildItem "\\172.16.205.123\nucleus-central\$you\$today\calls\"
```

**Expected:** you should see \*\_mic.wav, \*\_speaker.wav, \*.json, \*\_transcript.md.

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**Setup complete.** Tell Titu you're done.

### HELP

## If something fails

### Daemon log:

```
Get-Content "$NN\logs\voice_daemon.log" -Tail 50
```

scripts disabled on this system in **Step 6 or 7**:

```
powershell.exe -ExecutionPolicy Bypass -File  
"$NN\scripts\register-voice-daemon-task.ps1"
```

Test-Path **returned False in Step 5**:

```
ping 172.16.205.123  
cmdkey /list:172.16.205.123
```

If cmdkey doesn't show a nucleus user, re-run Step 5 in a **non-admin** PowerShell.

### Mic missing from recordings:

Teams → Settings → Devices → set Microphone to your Windows default input.

### UPDATE

## Update later

Run in **PowerShell** (re-set \$NN first if it's a fresh session):

```
Set-Location $NN  
git pull  
python -m pip install -r requirements.txt  
Stop-ScheduledTask -TaskName 'NAPCO Nucleus - Voice Daemon'  
Start-ScheduledTask -TaskName 'NAPCO Nucleus - Voice Daemon'
```

(Or just ask Titu to push the update remotely.)

### UNINSTALL

## Uninstall

Run in **PowerShell** (re-set \$NN first if it's a fresh session):

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```
Set-Location $NN  
.\scripts\register-voice-daemon-task.ps1 -Unregister  
.\scripts\register-chat-push-task.ps1 -Unregister  
cmdkey /delete:172.16.205.123
```

### CONTACT

## Contact

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Titu — [khasan@ael-bd.com](mailto:khasan@ael-bd.com)

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### APPENDIX

## Titu's remote-ops cheat sheet

*Not for the dev to run. Reference for managing dev PCs remotely once they've completed Step 8.*

### APPENDIX · A.1

## One-time setup on Titu's PC

Already done on .71. For reference if Titu ever reinstalls — **admin PowerShell**:

```
Start-Service WinRM
Set-Service WinRM -StartupType Automatic
Set-Item WSMAN:\localhost\Client\TrustedHosts -Value '172.16.205.*' -Force
```

### APPENDIX · A.2

## Credential header

Paste at the top of any new PowerShell session before running the commands below:

```
$pwd_at = ConvertTo-SecureString '606549' -AsPlainText -Force
$cred_at = New-Object PSCredential('AEL\khasan', $pwd_at)
```

### APPENDIX · A.3

## Dev PC IP registry (fill in as you onboard)

| Dev          | IP             | NUCLEUS_DEV_NAME | Repo path                 |
|--------------|----------------|------------------|---------------------------|
| Titu (yours) | 172.16.205.71  | Titu             | E:\Projects\NAPCO-Nucleus |
| Atik         | 172.16.205.108 | Atik             | F:\Titu vai\napco-nucleus |
| Rocky        | ?              | Rocky            | ?                         |
| Ferdows      | ?              | Ferdows          | ?                         |
| Amin         | ?              | Amin             | ?                         |
| Isruk        | ?              | Isruk            | ?                         |
| Assad        | ?              | Assad            | ?                         |

## APPENDIX · A.4

## Remote operations

Replace <IP> with the target PC's IP from the table above. Replace <repo> with the dev's repo path.

**Probe a dev PC (sanity check):**

```
Invoke-Command -ComputerName <IP> -Credential $cred_at -ScriptBlock {
hostname; whoami; "OS: $((Get-CimInstance Win32_OperatingSystem).Caption)"
}
```

**Tail their voice daemon log:**

```
Invoke-Command -ComputerName <IP> -Credential $cred_at -ScriptBlock {
Get-Content "<repo>\logs\voice_daemon.log" -Tail 30
}
```

**Check Scheduled Task state** (schtasks — Get-ScheduledTask needs admin remotely):

```
Invoke-Command -ComputerName <IP> -Credential $cred_at -ScriptBlock {
schtasks /query /tn "NAPCO Nucleus - Voice Daemon" /fo LIST
schtasks /query /tn "NAPCO Nucleus - Chat Push (Day)" /fo LIST
}
```

**Apply git pull + restart their daemon:**

```
Invoke-Command -ComputerName <IP> -Credential $cred_at -ScriptBlock {
Set-Location "<repo>"
git pull
Stop-ScheduledTask -TaskName 'NAPCO Nucleus - Voice Daemon'
Start-ScheduledTask -TaskName 'NAPCO Nucleus - Voice Daemon'
}
```

**Recover stuck WAVs to central** (when the dev's cmdkey wasn't set up correctly — cmdkey can't be done via WinRM, but New-PSDrive with explicit cred works):

```
Invoke-Command -ComputerName <IP> -Credential $cred_at -ScriptBlock {
$smb_pwd = ConvertTo-SecureString 'E7CqJ0d1oHox7HTjxNp_osD_fSyUe59I' -AsPlainText
-Force
$smb_cred = New-Object PSredential('nucleus', $smb_pwd)
New-PSDrive -Name NN -PSProvider FileSystem -Root '\\172.16.205.123\nucleus-central'
-Credential $smb_cred | Out-Null
$today = Get-Date -Format "yyyy-MM-dd"
$dest = "NN:\<DEV_NAME>\$today\calls" # replace <DEV_NAME>
New-Item -ItemType Directory -Force -Path $dest -ErrorAction SilentlyContinue | Out-Null
Get-ChildItem '<repo>\data\teams\calls\*' -ErrorAction SilentlyContinue | Copy-Item
-Destination $dest -Force
Get-ChildItem $dest | Select-Object Name, Length
Remove-PSDrive NN
}
```

**Open an interactive remote shell** (good for debugging):

```
Enter-PSsession -ComputerName <IP> -Credential $cred_at
# prompt becomes [<IP>]: PS> -- type commands; runs on their PC
# 'exit' when done
```

## APPENDIX · A.5

### Central host (.123) operations

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```
ssh ubuntu@172.16.205.123 # password: ayusuf

# Stack health
cd /home/ubuntu/napco-nucleus/deploy/linux-central
./status.sh

# Trigger daily-draft on demand (instead of waiting for BD 23:45)
docker compose exec daily-draft python collect_central.py --client all --last-minutes
1440

# Safe redeploy (after pushing a fix to main)
./deploy.sh # core stack only
./deploy.sh --runner # also recreates the GHA runner container

# Tail one worker
docker compose logs -f --tail 100 transcribe
```

## APPENDIX · A.6

### Daily-draft email destination

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Drafts land in khasan@ael-bd.com Gmail Drafts folder (configured via VERIFICATION\_TO in .env on .123). Subject pattern: Requirements Verification - YYYY-MM-DD.